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Tunnelling challenges on Singapore's 43km-long Thomson-East Coast Line

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ABSTRACT: Since 1995, the Land Transport Authority (LTA) has been developing Singapore's Mass Rapid Transit (MRT) system. The Thomson-East Coast Line (TEL) is the sixth MRT line, spanning from the north of Singapore to the city centre and extending along the east coast to the airport. This fully underground line spans over 43km. Civil works commenced in 2014 in the northern section, and the line's opening is planned in 5 stages, between 2020 and 2026. This paper describes how specific engineering challenges on TEL in varying geology were successfully overcome through a combination of customized TBM design solutions such as the use of dual crushers, screw conveyor pipe extensions; and through specialized ground treatment techniques such as ground freezing.

1 INTRODUCTION

At 43km, the Thomson-East Coast Line is one of the longest driverless rapid transit lines globally. The sixth addition to the Singapore's MRT network, it serves 32 stations with 7 interchange stations connecting to existing metro lines. It follows a route that combines both

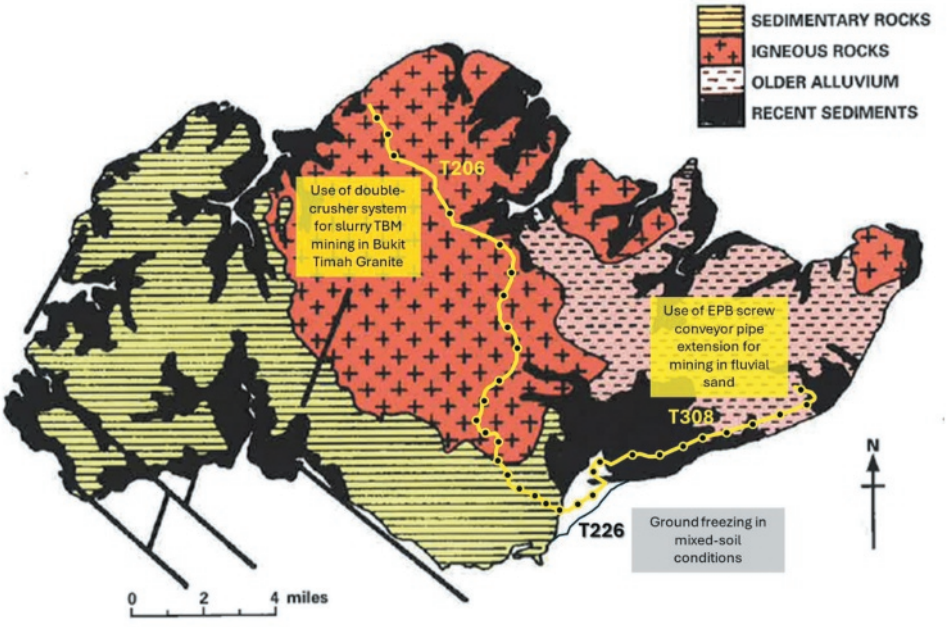


Figure 1. The Thomson-East Coast Line alignment set against Singapore's geological profile, with the three projects that are profiled in this paper.

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north-south and east-west corridors, commencing in the northand then proceeding south to the city center. Subsequently, it extends eastward along the country's eastern coast with plan of connection to the future Changi Airport Terminal 5, encompassing Kallang, Marine Parade, southern Bedok, and concluding at Upper East Coast.

Despite Singapore's limited land area, it possesses a well-documented and diverse geological profile. The north and central zones of Singapore are characterized by hard rock, while the southern and eastern parts of Singapore predominantly comprise alluvial and fluvial deposits.

Figure 1 overlays the Thomson-East Coast Line alignment against Singapore's geological profile. Tunnelling for the Thomson-East Coast Line has been successfully completed using 52 Tunnel Boring Machines (TBM), including 28 Slurry TBM used in the central zone in rock and 24 Earth Pressure Balance (EPB) TBMs used when tunnelling in soft rock, alluvial and recent fluvial deposits.

This paper delves into how selected projects had overcome challenging ground conditions through a combination of TBM selection and engineering innovation.

2 OVERCOMING GROUND CHALLENGES THROUGH TBM DESIGN

2.1 *Tunnelling in Bukit Timah Granite formation on contract T206*

Contract T206 comprises 6.2km of tunnels across ten drives. Five tunnel boring machines (TBMs) were employed: two slurry and three earth pressure balance types. This section examines three of the tunnel drives that had encountered hard rock at depths of between 24m and 29m below ground level. On these drives, 80% of the alignment was in Bukit Timah Granite of grade I (GI) and grade II (GII), which is extremely strong and intact in nature.

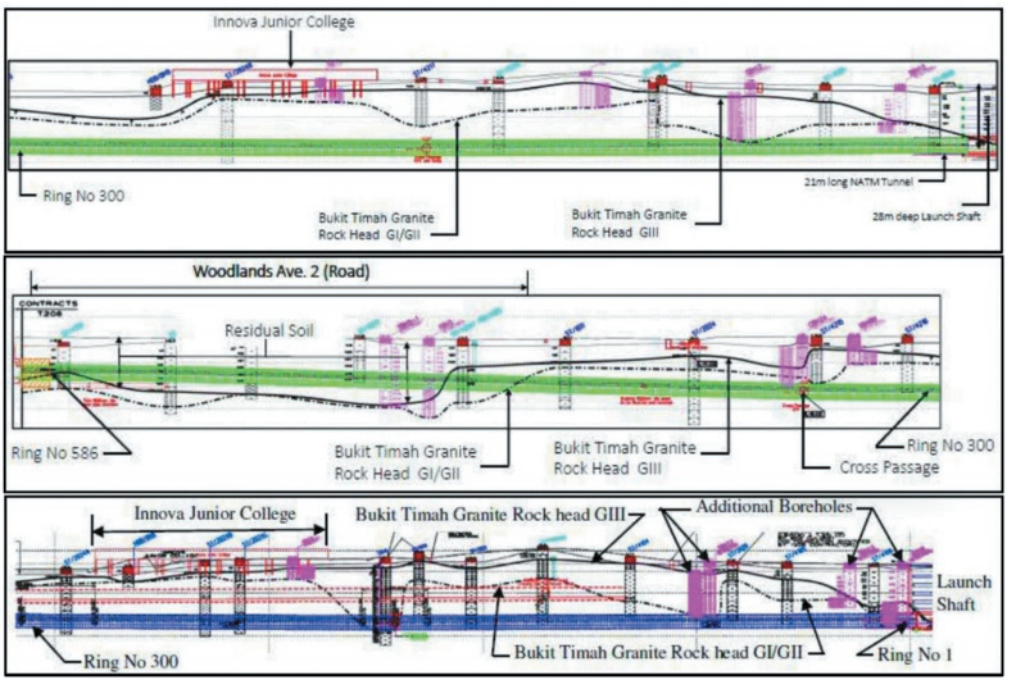


Figure 2. Geological profile of the three drives with an approximate 80% in GI and GII.

Pre-tender and additional boreholes had ascertained samples of GI and GII ranging from unconfined compressive strength (UCS) of 130MPa to 268MPa. In addition, the CERCHAR index of these samples varied from 4 to 5.5, confirming the granite encountered to be strong/very strong and extremely abrasive in nature. Figure 2 shows the geological profiles of these three drives.

Due to the geology of the tunnel drives, a slurry shield TBM was specified in the contract. The cutterhead was equipped with 19-inch disc cutters, with a total of 43 disc cutters fitted, including 6 gauge cutters. Additionally, 54 scrapers were installed on the cutterhead. The spacing of the disc cutter tracks was maintained at 80mm, and the opening ratio was kept at 19%. This arrangement of tools on the cutterhead enhanced the efficiency of crushing granite rock and facilitated chip formation. Figure 3 shows the slurry shield TBM specifications with the cutterhead design.

Slurry Shield TBM Specifications	
Outer Diameter	6.670m
Cutterhead Tools	19" Disc Cutters, 80mm track spacing (43 in total – 6 Gauge Cutters, 29 Face Discs, 8 Centre Cutters) 54 Scrapers
Thrust Jacks	22 jacks with maximum stroke of 2200mm. Total maximum thrust of 41,800kN.
Cutterhead Motors	9 x 160kW (1,440kW in total)
Cutterhead Speed	Maximum: 7rpm
Articulation Jacks	12 pcs
Tail Seal Brushes	3 rows of tail seal brushes

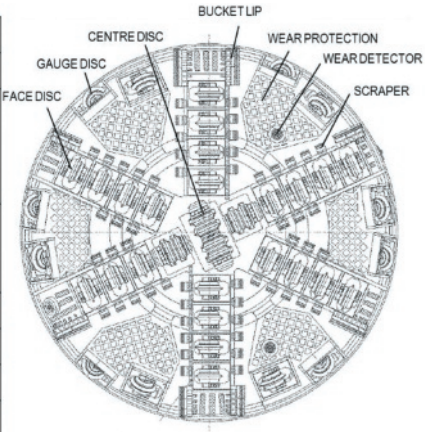


Figure 3. Cutterhead design of slurry shield TBMs on T206.

The TBM was equipped with a jaw crusher and a roller crusher, forming a double-crusher system. The jaw crusher, located behind the submerged wall, primarily functioned to reduce large granite pieces to sizes of approximately 150mm. The roller crusher, situated on the first backup gantry just behind the segment erector, performed secondary crushing, reducing granite to chips of 40mm to 50mm. The outgoing flow of slurry to the Slurry Treatment Plant (STP) through the slurry line was approximately 450m³. After passing through the roller crusher, the flow entered a distributor. The distributor's function was to facilitate the recirculation of slurry back to the working chamber. A schematic of the double-crusher system and the slurry circuit is shown in Figure 4.

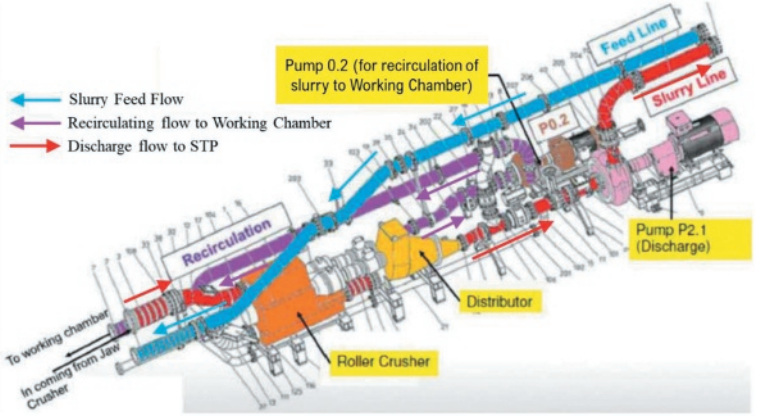


Figure 4. Schematic of double-crusher system and slurry circuit.

Approximately 30% of the flow from the discharge line was recirculated to the working chamber. This recirculation reduced the flow to the STP, consequently allowing for a smaller discharge line diameter in the TBM and downsized equipment in the STP. The reduction in

equipment sizes contributed to energy savings. Additionally, the double-crusher system helped reduce blockages in discharge lines, resulting in fewer maintenance stoppages. Moreover, the granite chips (at 40-50mm) were suitable for recycled use as road base.

During TBM operation, cutterhead interventions (CHIs) were performed at 10-ring intervals to measure cutter-wear. While most such interventions only required the retightening of nuts and bolts for the disc cutters, cutters on the outer periphery of the cutterhead showing wear of 5-10mm were interchanged with disc cutters on face to maximise economy. Combined with the abovementioned double-crusher system, the average rate of mining was approximately 5.6m/day in full face granite, with a total of 30 rings (i.e. 42m) built in the best performing week.

2.2 *Tunnelling in fluvial sand on T308*

Contract T308 is in the densely populated Marine Parade area of Singapore. The 1.34km twin tunnels run beneath Marine Parade Road, between Marine Crescent Road and Marine Terrace. The surrounding area includes a mix of private condominiums, public residential estates, schools, and various amenities. The geological profile for T308 involves the TBMs tunnelling through the Kallang Formation, which includes sections of fluvial sand and marine clay, before transitioning into Old Alluvium towards the end of the drive. The soil overburden along the tunnel alignment ranges from 13m to 28m from the surface to the tunnel crown (see Figure 5).

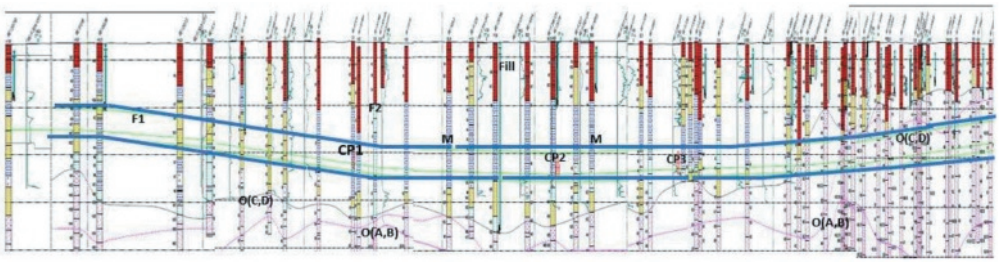


Figure 5. Geological profile along T308 tunnel alignment.

Further assessment of the ground conditions indicated that 25% of the entire tunnel length would be through fluvial sand. As fluvial sand is cohesionless and highly permeable, tunnelling through full-face sand presented a major challenge for T308, particularly in terms of controlling face pressure support and minimising ground settlement.

Grouting the soft soils before tunnelling would have been the ideal mitigation measure. However, this would have necessitated extensive road diversions, resulting in significant inconvenience to stakeholders. As a compromise, grouting works were limited to locations planned for TBM stoppage and cross passages, affecting only small road sections.

Two Earth Pressure Balance (EPB) TBMs machines were used on this project. While EPB TBMs are well-suited for soft ground conditions, the loose fluvial sand composition hinders plug formation in the plenum chamber behind the TBM cutterhead. This negatively impacts the TBM's ability to maintain face pressure support during tunnelling, potentially resulting in excessive settlement or even ground loss. Consequently, this poses an immediate risk when tunnelling beneath Marine Parade Road and adjacent structures.

Specific requirements on TBM design were incorporated at tender, such as equipping the TBMs with a factory-fitted secondary screw gate and four rows of tail seal brushes. In addition, the T308 contractor collaborated with the TBM manufacturer to customize the provision of a long discharge pipe from the discharge end of the screw conveyor. The discharge pipe had an approximate diameter of 400mm and was 30m long, comprising a combination of steel pipe and heavy-duty flexible hoses. It connected from the screw conveyor to the second-last backup gantry. The pipe included two automatic valves, equipped with manual override, that could close immediately in case of soil ingress or power failure. Additional gate valves and a pneumatic butterfly valve along the discharge pipe, as shown in Figures 6 and 7, helped control pipe opening and regulate material discharge within the pipeline.

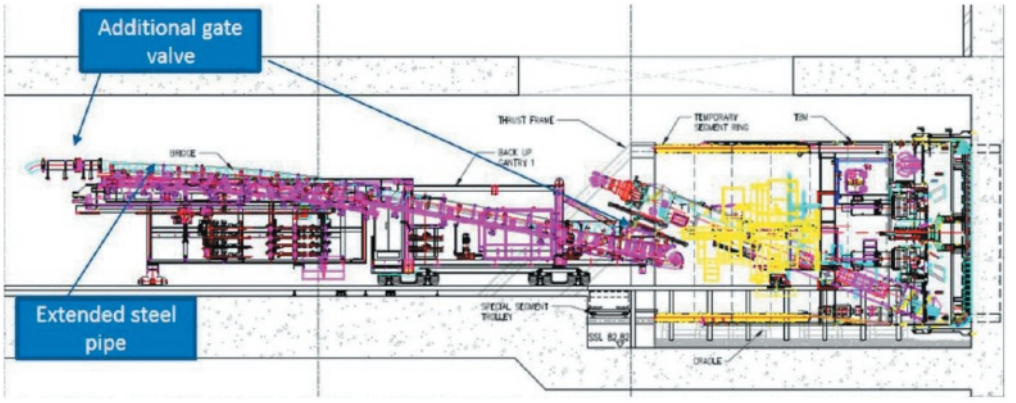


Figure 6. Schematic of EPB TBMs fitted with an extended discharge pipe.

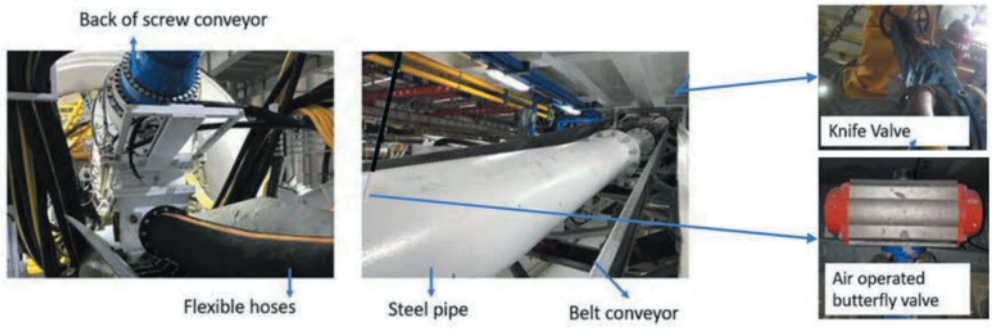


Figure 7. Long conveying steel pipe used to control pipe opening and regulate discharge of materials.

These measures significantly mitigated the risk of screw blow-out when tunnelling through F1 sections without ground improvement. The installation of this extended discharge pipe connected to the screw conveyor's outlet minimised muck spillage, enhanced muck accounting accuracy, and improved overall cleanliness. For the twin tunnel drives through the F1 ground, excavation management was maintained within 85% to 115% of the expected excavated weight and volume.

TBM mining speed averaged 25-30mm/min in Kallang Formation, with 4 rings/day (5.6m) productivity. In full face fluvial sand, increased thrust force and torque reduced speed to 5-10mm/min. Productivity dropped to 1-2 rings/day due to time spent clearing spoil blockages in the steel pipe near the screw gate. Ground movement induced by the twin bored tunnel drives remained within project limits. The first tunnel drive (Woodlands-bound) resulted in a 10mm settlement, while the combined effects after the second tunnel drive (Changi-bound) led to a total settlement of 35mm. The recorded settlements when tunnelling in fluvial sand were plotted, with the volume loss after the twin tunnel drives computed at 1.2%, lower than the contractual volume loss requirement of 3% in Kallang Formation. Upon completing the F1 sections, the extended conveying pipe arrangement was dismantled for the remaining tunnel drive.

3 OVERCOMING GROUND CHALLENGES THROUGH GROUND IMPROVEMENT

3.1 *Ground freezing for the construction of SCL tunnels on T226*

Ground freezing was used for the first time for a railway project in Singapore on Contract T226. The project involved integrating three MRT lines, namely the existing North South Line (NSL) and Circle Line (CCL), and the new Thomson-East Coast Line (TEL) (see Figure 8).

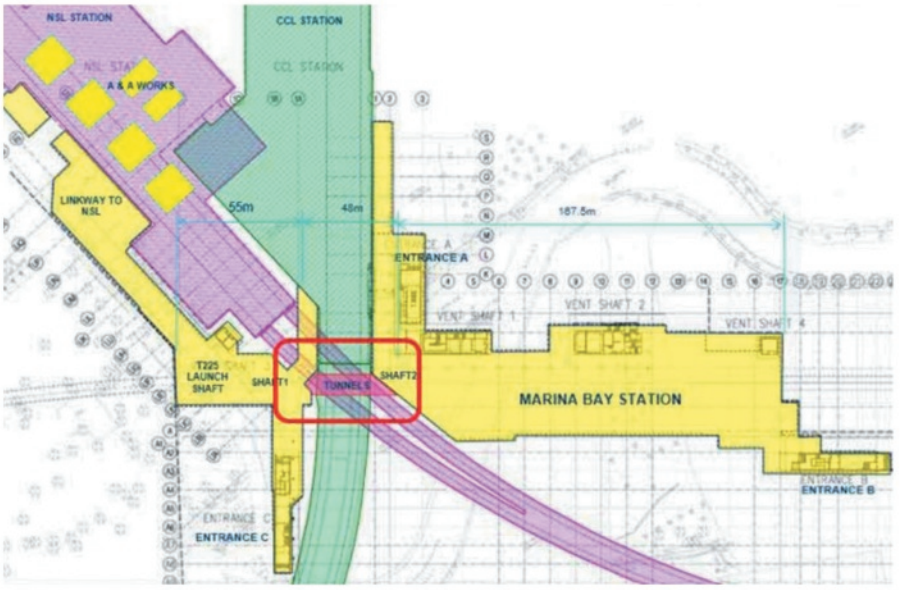


Figure 8. Plan view of the intersecting NSL, CCL and TEL tunnels.

The TEL tunnel alignment was in mixed-soil condition, comprising hydraulically reclaimed Sand Fill overlying the Kallang Formation. The Kallang Formation includes Upper Marine Clay (UMC), Estuarine Clay (E), Fluvial Sand (F1), Fluvial Clay (F2), and Lower Marine Clay (LMC). Beneath this lies the Old Alluvium (OA), which exhibits various weathering grades: OA(E) (Residual soil, N<10), OA(D) (Destructured, 10<N<30), OA(C) (Distinctly weathered, 30<N<50), OA(B) (Partially weathered, N>50), and OA(A) (Unweathered, N>100), where N refers to the Standard Penetration Test (SPT) N-value. The soil layers varied significantly throughout the site, as may be seen from the two boreholes within this location as shown in Figure 9.

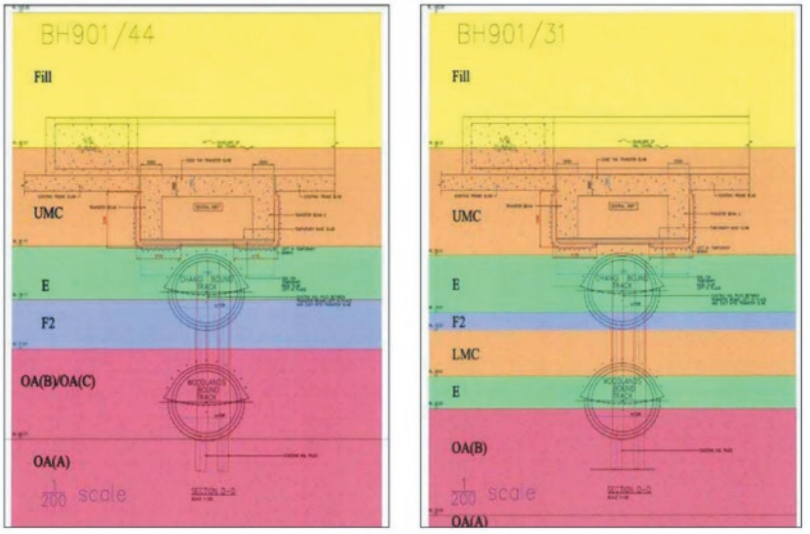


Figure 9. Soil profile for the twin TEL tunnels to be built.

The two main tunnels of the TEL undercross the NSL and CCL tunnels. Since the NSL tunnels are supported on bored piles, TEL construction required the removal of some of these

piles to make way for the new tunnels. For this short stretch of tunnels, the construction of a Sprayed Concrete Lining (SCL) tunnel was the most viable option (see Figure 10).

The SCL method requires dry conditions for stability. The upper tunnel soil, comprising LMC, was improved using Jet Grouting (JGP). The lower tunnel, however, comprised a mix of LMC and OA, complicating soil improvement. While JGP was used for LMC, it proved unreliable for denser OA grades, and chemical grouting had limited historical success. This necessitated an innovative solution: Ground Freezing (GF). The GF objective was to create two frozen soil walls as impermeable layers, preventing water ingress and ensuring dry conditions for SCL tunnelling.

Once the GF system was activated and frozen soil walls verified by temperature sensors, SCL mining of the lower tunnel commenced. After lower tunnel completion, GF was deactivated, followed by upper tunnel construction. SCL construction involved sequential excavation and temporary shotcrete lining installation for top heading, bench, and invert. The tunnels, located directly under the Linkway, required the removal of 16 partially removed NSL pile remains from the Linkway construction.

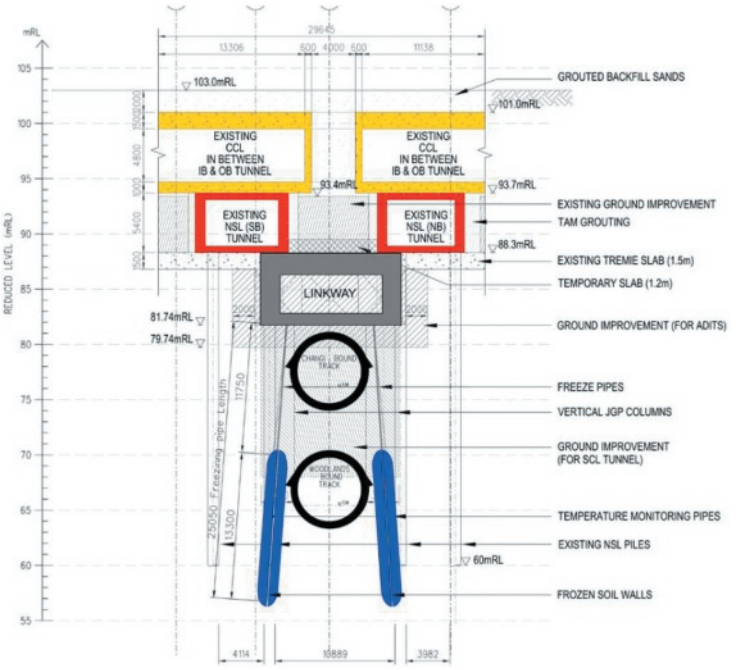


Figure 10. Cross section of SCL tunnels and frozen soil walls.

Ground Freezing (GF) offers significant advantages over conventional grouting techniques. It is widely applicable across various soil types and minimally invasive, making it ideal for constrained sites. GF effectively controls water movement while also enhancing soil strength properties. Its effectiveness relies on the presence of free water in the targeted zone, which is typically abundant in problematic areas. However, GF implementation requires careful consideration. The method's reliance on water presents a paradox: water is both the problem to address and essential for the freezing process. In dry conditions, GF may be ineffective, though water ingress is no longer a concern under such circumstances. Engineers must also account for potential soil expansion during freezing and settlement during thawing. Calcium chloride ($CaCl_2$) brine was the GF agent of choice on T226.

T226's primary design requirement was to form two impermeable walls to facilitate SCL tunnel mining. The design focused on wall impermeability rather than capitalising on the increased strength of frozen soil. The main challenge was in determining the optimal size and

spacing of frozen soil columns to achieve continuous walls efficiently, balancing considerations such as the number of freezing pipes, time required to achieve target wall thickness and column size/spacing.

The optimal design resulted in frozen soil columns of 1.8m diameter, spaced at 0.9m centre-to-centre. This large overlap, 0.5 times the diameter, ensured wall continuity and watertightness while allowing for alignment inaccuracies. In areas where existing structures interfered with the proposed frozen soil walls, the freezing pipe spacing was realigned to form walls that wrapped around these structures. This approach allowed the design to adapt to the site's specific conditions whilst maintaining the effectiveness of the ground freezing system (see Figure 11).

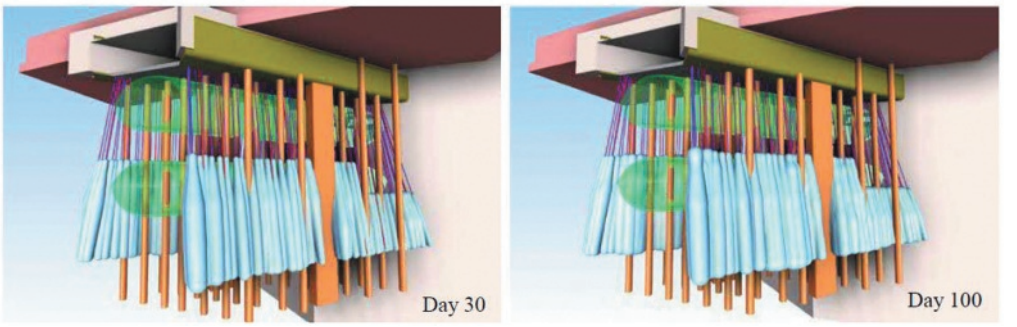


Figure 11. Development of brine-frozen soil walls using BIM based on temperature monitoring results.

GF had raised initial concerns about soil expansion, but actual results differed from predictions. Vertical displacement measurements of the North-South Line (NSL) tunnels, obtained through the Automated Tunnel Monitoring System (ATMS), showed significantly less structural deformation than anticipated. The expected heaving due to frozen soil expansion was also not observed.

4 CONCLUSION

Tunnelling for the Thomson-East Coast Line has been successfully completed, overcoming challenges related to the urban environment, the complex geological conditions and tight completion schedule. The three case studies presented in this paper demonstrate innovative solutions to diverse ground challenges in underground construction. The double-crusher system proved effective in hard Bukit Timah Granite, accelerating mining progress. For tunnelling in fluvial sand, the extended discharge pipe minimised muck spillage and improved settlement control. The pioneering use of ground freezing successfully mitigated water ingress in mixed soil conditions. As Singapore continues to expand its railway network, these valuable lessons and technological advancements will significantly enhance industry expertise, paving the way for more efficient and innovative underground construction practices. These experiences not only contribute to local knowledge but also position Singapore as a leader in urban underground development, offering insights that may benefit similar projects globally.

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